Investigating The Effects of Card Shuffling Practices

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Executive Summary

When playing cards, the method of shuffling and the number of times the deck of cards is shuffled are responsible for the distribution of cards dealt in the following hand. A study was conducted on the effects of riffle and overhand shuffling techniques with the purpose of either informing fair competition between consecutive game hands or exploiting shuffling practices for a competitive edge.
A scoring system was used to assess post-game card decks that feature groupings of card runs and sets, as well as the natural frequency of such groupings in randomly generated card decks. Statistical evaluation of the two types of decks showed different score distributions making each uniquely identifiable. This allows for a competitor to observe any deck of cards and determine, with a relatively high level of confidence, whether it has been shuffled after its previous use.
Simulated post-game decks were repeatedly shuffled by different techniques until they displayed distributions that matched randomly generated decks, an indicator that a new game hand no longer features signs of the previous game. It was found that the riffle shuffle technique is significantly more effective than an overhand method, requiring only 4-5 shuffles compared with >15 shuffles, respectively.

Quantitative Skills/Knowledge Demonstrated

This project deploys statistical theory to analyse card distributions, including use of the Central Limit Theorem, Bayesian/Conditional statistics, randomisation, convergence, and Normal/Gaussian distributions. An empirical approach is used, generating datasets with the use of Python and its typical packages (i.e., NumPy, Pandas, Matplotlib, SciPy, Statistics and Random) in a Microsoft Visual Studio and OS environment. Real-world mechanisms are translated into mathematical equivalents and subsequently constructed as computer algorithms/functions for empirical use. Results are graphed and discussed.

Introduction

Several motivations exist for the investigation into shuffling impacts on card distributions within game decks. Gaining a competitive advantage is an obvious motive, however the potential to silence that one relative at family gatherings that insists on uttering the phrase "well who shuffled this deck?" (implying poor shuffling upon viewing their newly dealt hand) is equally a strongly rewarding motive. This highlights the primary questions and objectives of this investigation: "Can you objectively tell the difference between a post-game deck and a random deck?", and "What shuffling method and how many shuffles are necessary for the deck to truly be objectively random again?". To answer these questions, an empirical study is conducted via computer simulation of decks and shuffling mechanics, utilizing the software language Python. The results are graphed and discussed, including highlighting objective conclusions and any potential further investigations.

Method

The following investigative steps will be taken to address the aforementioned questions:

1. Develop a computational representation of a card deck, a 'game hand', card sets, card runs, and a scoring system for recognizing the presence and location of sets/runs within a generated deck/hand.

2. Randomize the order of the card deck and investigate its numerical properties, namely the natural prevalence of sets/runs and their groupings within the deck. Additionally, investigate the number of decks which must be simulated to ensure repeatability of results/findings.

3. Develop a method to simulate 'post-game hands' and a respective 'post-game' card deck. Investigate its numerical properties and how they differ to randomised decks.

4. Identify shuffling techniques and develop their computational/numerical equivalence. With the techniques, investigate the effect of shuffling on established 'post-game' card decks.

Analysis

1) Development of model

A rather straightforward numerical model for a card deck is developed by generated a linear set of integers from 1 to 52. It can be assumed that each suite has 13 numbers each (e.g., 1-13, 14-26, 27-39, 40-52), and cards range from an aces as the smallest to a king as the largest. For example, the hearts suite is represented as (noting "oH" stands for 'of hearts'): AoH=1, 2oH=2, 3oH=3, 4oH=4, 5oH=5, 6oH=6, 7oH=7, 8oH=8, 9oH=9, 10oH=10, JoH=11, QoH=12, KoH=13.
Sets and runs of cards, and of various lengths, can then be identified and stored as known number pairs within a programmed library/list. For example, a four-card run (a term to denote suited cards in consecutive order) of hearts might be [JoH, QoH, KoH, AoH] which is identified as [11, 12, 13, 1]. Importantly, only the combination of numbers is necessary to identify this run, not the order in which they appear (i.e., [12, 13, 1, 11] is the same as [11, 12, 13, 1]). Similarly, a 3-card set (a term to denote cards of the same numeric, not suited) of aces might be [AoH, AoD, AoC] = [1, 14, 27].
For simplicity, the investigation will assume that the game being played is rummy. This game involves players attempting to make a 4-card set or run and a 3-card alternative set/run to win with a 7-card hand. Therefore, a numerical library is created of all 1-, 2-, 3- and 4-card runs/sets possible with a single 52-card deck. This approach is expected to yield the same results as investigating a canasta game, which simply doubles the card number, neglects runs, and changes the roles of aces, 2's and 3's.
At the end of a rummy game, we expect to see one winner and several other 7-card hands which are varying degrees towards completing a 3- and 4-card combination. Typical card play behaviour is for all hands to be thrown onto the discard pile before rejoining the discard pile with the unused card deck. Then, shuffling occurs.
A scoring system is introduced to aid identifying the prevalence of runs and sets in a given deck. 2-card gives 1 point, 3-card gives 2 points, and 4-card gives 3 points. Given a deck of cards, a sliding window algorithm is applied to score the deck: The algorithm looks the first seven cards in the deck (the typical hand size of rummy) and checks whether within that seven cards exists any known 2-, 3-, or 4-card set/run. It updates the deck score and proceeds to slide one card further into the deck and repeat its search (now looking at card positions 2 to 8). This means that 45 sets of 7 cards are investigated in each deck, all adding to a single score for the whole deck. Note that if a window of seven cards has a 4-card combination, it also has four 3-card and six 2-card sets present - this yields 17 points from a single 7-card window. A deck's total score can add up quickly.

2) Numerical properties of a typical random deck

The first step in the investigation is to evaluate the natural frequency of card/set combinations within randomly generated decks, and what their respective typical scores are given our chosen scoring system. Generating a random deck of cards is relatively easy: a computer algorithm is used to create a linear array of integers from 1 to 52 (with no numbers repeating) and then a randomization algorithm can be used to shuffle the order of these cards randomly.
There are 52! ($\sim 8.07*10^{67}$)(here, "!" is the mathematical notations for 'factorial') ways to arrange the cards of a 52-card deck. Simulating all of these is unreasonable and computationally expensive. Thankfully, we care more about the combinations of seven cards that exist (which is much less than 52!): this is denoted in combinatorics as "52C7" ("52 choose 7") and is calculated as approximately 134 million possible 7-card combinations. This remains unreasonably large of a simple study and therefore a convergence analysis will be used to reduce the required number of simulations.
To conduct a convergence analysis we simulate a random card deck, score it, and store its score in memory. The step is repeated and then an average score for the two decks is calculated. We continue to generate new card decks, score them and evaluate the new cumulative average score. For each new deck, a convergence analysis is interested in how much the calculated average score is changing due to a new deck having been added to the calculation. When we observe that the average score is no longer changing a significant amount, we can assert

that we have generated a reasonable proportion/sample of card decks to represent all possible card decks. This is the same as saying that "we don't need to generate more and more card decks beyond 'x'-number because we no longer expect the result we're calculating to change significantly".
Figure 1 shows the results of the convergence analysis plotted on a log-log graph for ease of viewing various magnitudinal orders. Here, one thousand random decks were simulated and scored, calculating the average score for each new deck added to the convergence study. The graph shows that once a thousand decks are simulated, the average is expected to change less than 0.01%. It takes the computer less than 30 seconds to generate this many decks and score them, which is ideal for keeping this study simple while remaining accurate.

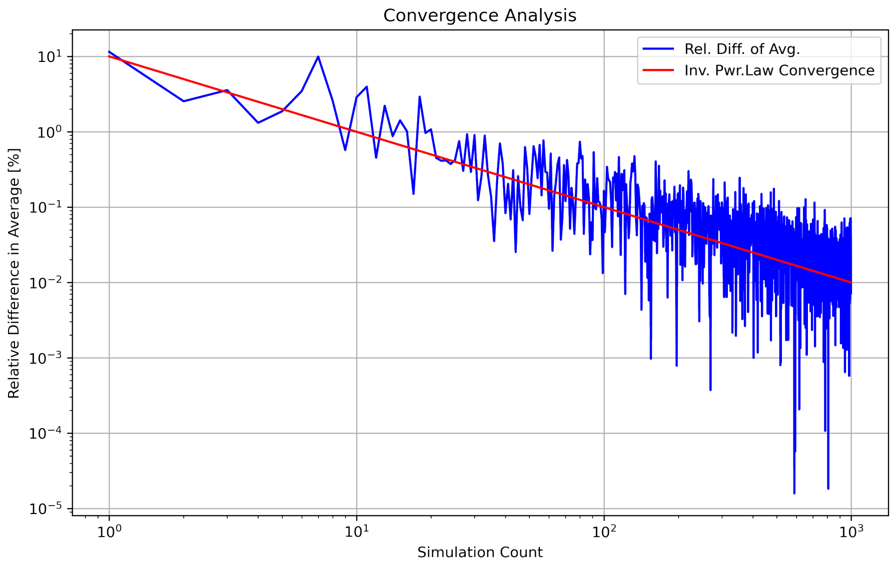

Figure 1: Convergence analysis of average random deck score (blue) with observed power trend (red line)

We can now analyse the average and general distribution of scores observed in the randomly generated decks. Figure 2 shows a thousand randomly generated card decks and their respective scores, alongside the moving average score calculation which underpinned the convergence analysis above. The Central Limit Theorem is a useful distribution law which observes that with increasing sample/simulation size, the distribution of resulting scores will trend towards what is known as a normal distribution. We therefore fit a normal distribution to the scores observed so that a mathematical description can be provided. Figure 3 shows the respective histogram plot and fitted normal distribution curves. These results state that a random deck of cards has an average score (μ) of 108 with a standard deviation (σ) of 26 (denoted $X_{random} \sim \mathcal{N}(\mu = 108, \sigma = 26)$.

3) Numerical properties of a 'post-game' deck

A similar investigation of post-game decks can be conducted. Assuming 3-4 players, random card/set combinations of differing completeness are generated for all players and then assorted into a randomized deck of remaining cards at different positions. The simulated 'post-game' decks are then scored and the distribution of scores is investigated.
Figure 4 shows the resultant distribution of a thousand post-game decks, demonstrating an average score (μ) of 278 with a standard deviation of (σ) of 46 (denoted $X_{postgame} \sim \mathcal{N}(\mu = 278, \sigma = 46)$. The two distributions can be compared to show their distinct differences, as shown in Figure 5. As a result, there is a high confidence that given a deck's score, it would be possible to assert whether the deck was random or post-game origin without having knowledge of its prior use. This is an excellent example of conditional probability (i.e., Bayesian statistics).

4) Shuffling techniques and their effectiveness

Finally, we're interested in how many times a post-game deck must be shuffled before it appears (scores) as a random deck again. This is of interest as this represents the deck no longer being influenced by the previous game

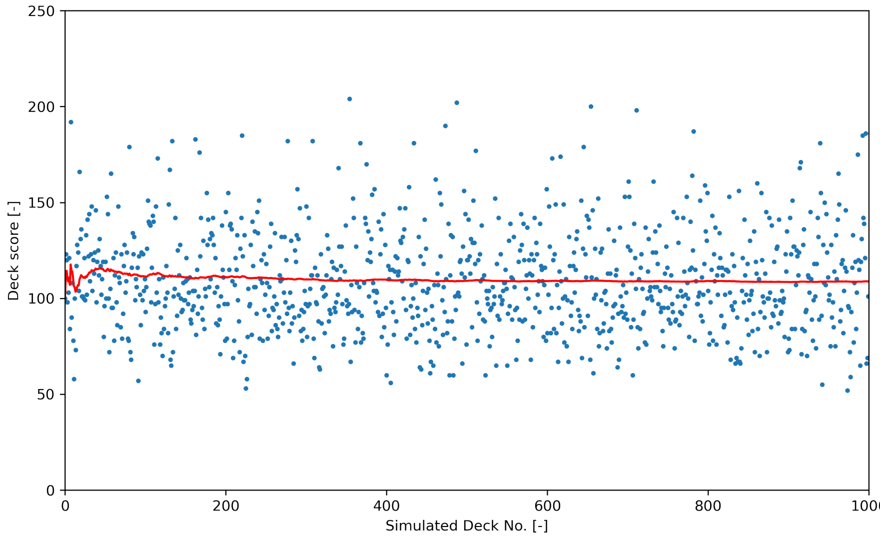

Figure 2: Analysis of random deck scores (blue dots) with moving average calculation (red line)

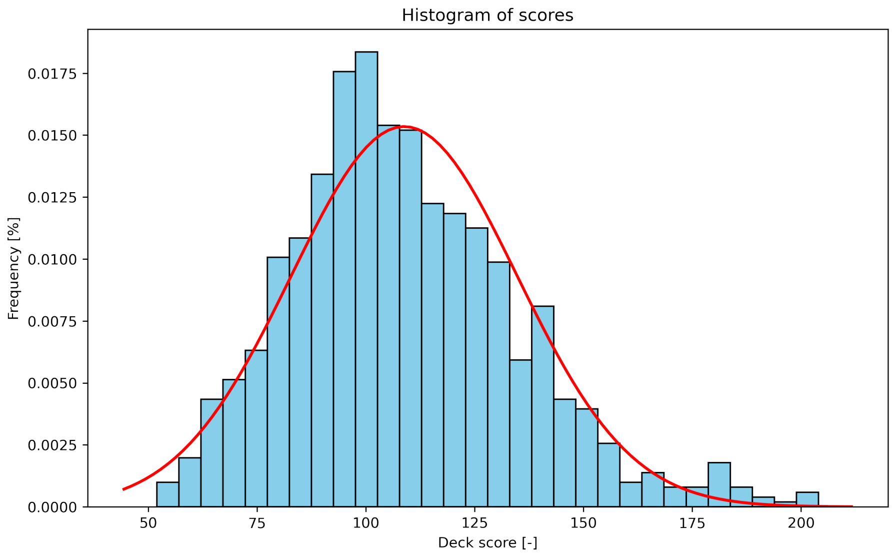

Figure 3: Score distribution of one thousand random simulated decks demonstrating normal distribution fitting

hand, meaning, should a player receive a card in their new hand which they'd possessed in the previous game, it isn't explicitly due to shuffling practises.

Two different shuffling techniques are considered and algorithmically simulated: the riffle shuffle and the overhand shuffle. The riffle shuffle splits the deck in approximately two halves and then proceeds to fan the two halves together so that a new deck is created through compiling alternative cards from each halve. The technique isn't exact, leading to roughly 1-4 card group alternations between halves, and the 'halves' themselves aren't exactly 26 cards either. The overhand shuffle takes a sizeable stack of a primary deck of cards (leaving a few to form the base on a new deck) and proceeds to slide roughly 1-4 cards, at a time, from the top of the stack onto the new deck. The mechanics of the two techniques mean they can roughly be described as a dispersion approach (the riffle shuffle) and a displacement approach (the overhand shuffle).

Figures 6 and 7 show the results of the analysis. For all thousand simulated post-game decks, each were shuffled via the chosen techniques and their scoring distributions reassessed for likeness to the random deck distribution after each shuffle. The results show that the riffle shuffle requires only 4-5 shuffles before the deck is indistinguishable from a random deck distribution, meanwhile the overhand shuffle requires in excess of 15 shuffles.

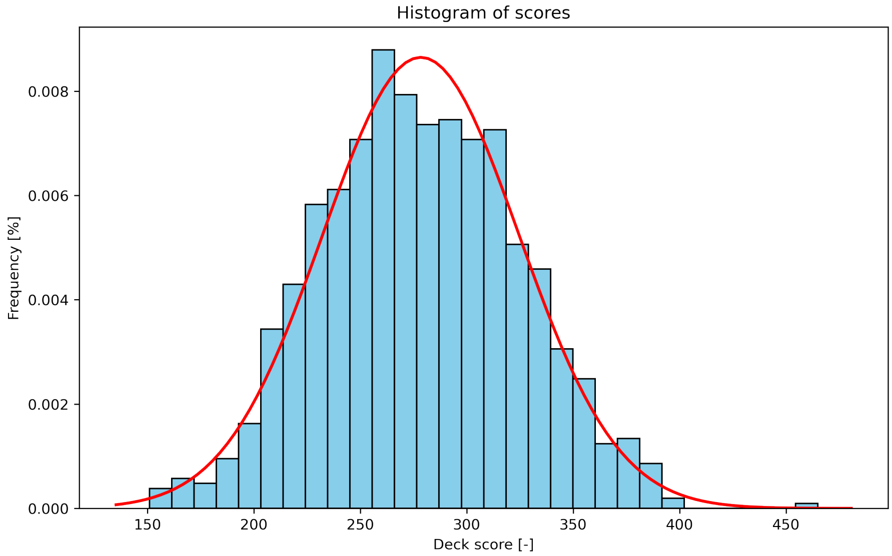

Figure 4: Score distribution of one thousand 'post-game' simulated decks demonstrating normal distribution fitting

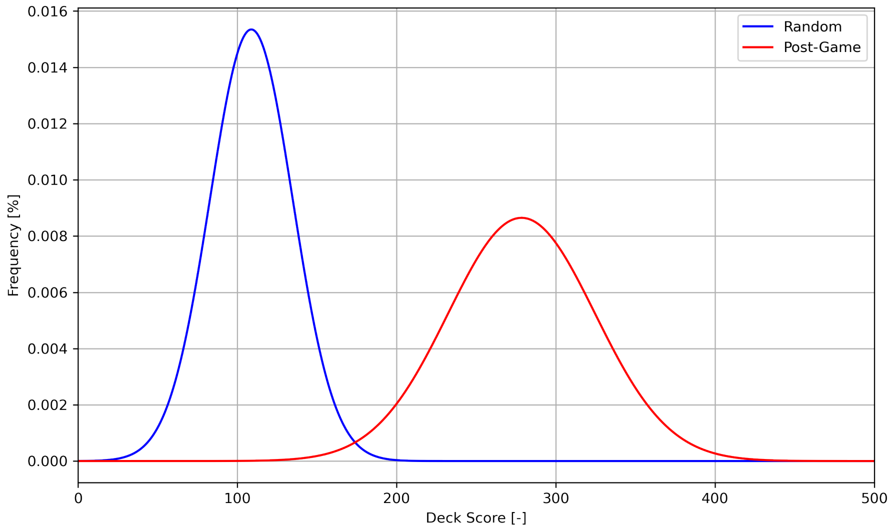

Figure 5: Comparison of random and post-game scoring distributions, showing distinctly different characteristics

Further Investigations

While the simulations investigated the effectiveness of the shuffles in returning post-game decks to a randomised likeness for fair competition, they stopped short of assessing the impact on play where a randomised state wasn't achieved. It therefore remains unanswered whether players will experience a discernable change in play whether the deck is shuffled sufficiently or not. Further investigation is required to determine whether a player can predictably receive a higher percentage of cards from their previous hand, if the deck isn't shuffled back to a random state.
It is hypothesised that the manner in which the post-game deck is recompiled (e.g., the order of hands, discard pile and remaining cards) and the use of the overhand technique (which displaces small groups of cards instead of dispersing them) would permit a potential competitive advantage if used or observed correctly.

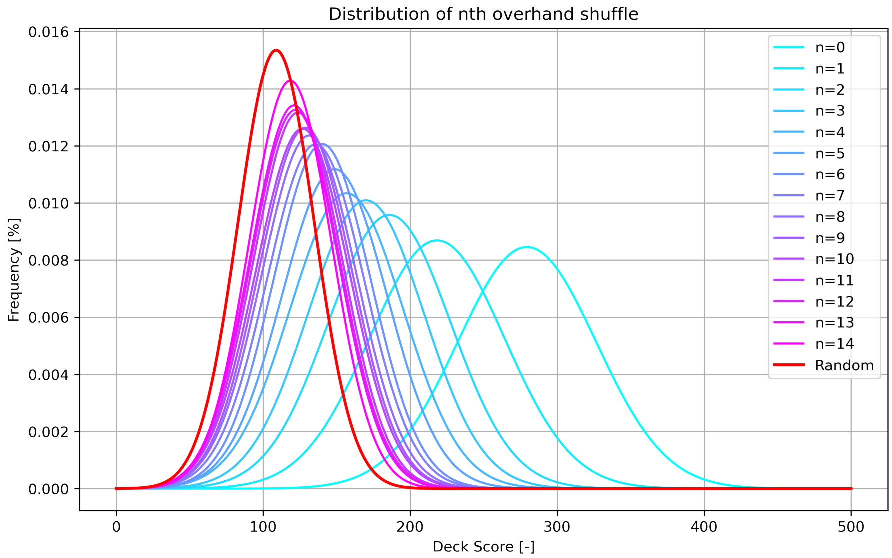

Figure 6: Affect of overhand shuffle on score distribution, showing approximately 15 shuffles are required to re-establish a randomized distribution. ("n" denotes number of shuffles)

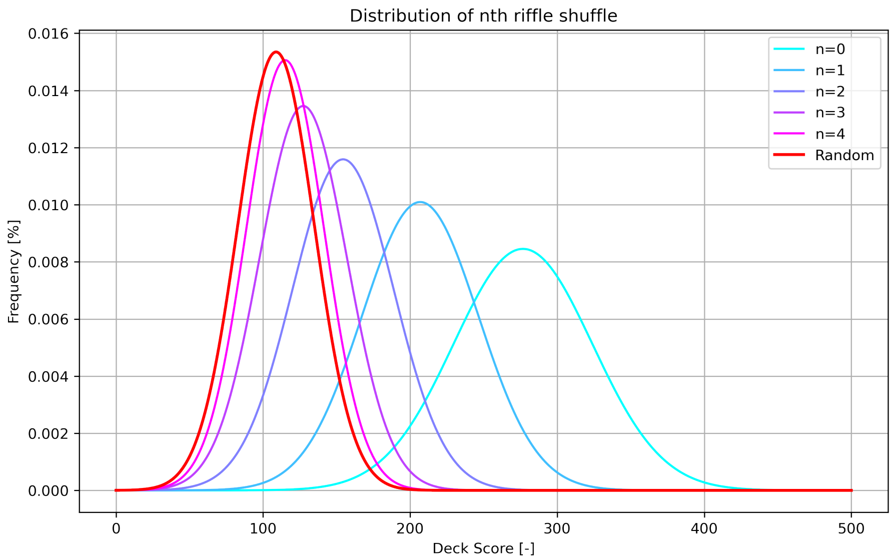

Figure 7: Affect of riffle shuffle on score distribution, showing approximately 4-5 shuffles are required to re-establish a randomized distribution. ("n" denotes number of shuffles)